

A cross-alphabet identity for bounded-difference square arrays: OEIS A125531

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14 September 2026

Abstract

OEIS A125531 carries an empirical identity relating its counts to those of the same problem over an alphabet one smaller. It is true, and sharply so: the threshold the entry states is exactly the diameter bound of the grid graph, and the identity fails below it. The proof is four lines and uses no machinery.

2020 Mathematics Subject Classification. 05A15, 05B45.

1 The conjecture

OEIS A125531 is “Number of $n \times n$ arrays with entries in $1..4$ in which adjacent entries differ by 1 or less (adjacent means in x or y directions).”. It has offset 0 and begins

1, 4, 54, 3682, 1209830, 1909172038, 14441437270426, 523308354987531254, ...

The entry states, as a conjecture contributed by R. H. Hardin and never marked settled:

[Empirical] $a(\text{base}+1, n, \text{diff}) = a(\text{base}, n, \text{diff}) + F(n, \text{diff})$ for $\text{base} \geq 2 \cdot \text{diff} \cdot (n-1)$ and $F(n, \text{diff}) =$
($n=2$, A063496($\text{diff}+1$)) ($n=3$, A068744($\text{diff}+1$)) ($n=4$, A068745($\text{diff}+1$)) ($n=5$, A068746($\text{diff}+1$))
($n=6$, A068747($\text{diff}+1$))

As of the “Last modified” line on the live entry (Mar 31 2012 12:35:05, revision 6) this is still recorded as empirical.

2 Notation

For integers $b \geq 1$, $d \geq 1$ and $n \geq 1$ write $T(b, d, n)$ for the number of $n \times n$ arrays with entries in $\{1, \dots, b\}$ in which any two orthogonally adjacent entries differ by at most d . The entry is $T(4, 1, n)$, and the conjecture is

$$T(b+1, d, n) = T(b, d, n) + F(n, d) \quad \text{for } b \geq 2d(n-1),$$

with $F(n, d)$ the quantity the entry tabulates through its reference sequences ($n = 2$: OEIS A063496, $n = 3$: OEIS A068744, $n = 4$: OEIS A068745, $n = 5$: OEIS A068746, $n = 6$: OEIS A068747).

3 The proof

Theorem 1. *For all $n \geq 1$, $d \geq 1$ and $b \geq 2d(n-1)$,*

$$T(b+1, d, n) - T(b, d, n) = F(n, d),$$

where $F(n, d)$ is the number of admissible arrays over $\mathbb{Z}$ whose maximum entry is 0 — a quantity depending on n and d only.

Proof. Step 1. An admissible array over $\{1, \dots, b+1\}$ either uses the value $b+1$ or does not, and those that do not are exactly the admissible arrays over $\{1, \dots, b\}$. So $T(b+1, d, n) - T(b, d, n)$ counts the admissible arrays over $\{1, \dots, b+1\}$ that use $b+1$.

Step 2. Every admissible array satisfies $\max - \min \leq 2d(n-1)$. Let u attain the maximum and v the minimum. In the grid graph on $n \times n$ cells with orthogonal adjacency any two cells are joined by a path of length at most $2(n-1)$ — move along a row, then along a column — and along a path of length L the entry changes by at most dL , since each step is between adjacent cells.

Step 3. An array counted in Step 1 has maximum exactly $b+1$, so by Step 2 all of its entries lie in $[b+1-2d(n-1), b+1]$. When $b \geq 2d(n-1)$ that interval is contained in $\{1, \dots, b+1\}$, so the constraint “entries in $\{1, \dots, b+1\}$ ” is implied by the maximum and the range bound and imposes nothing further. The arrays counted in Step 1 are therefore in bijection — by the translation carrying the maximum to 0 — with the admissible arrays over $\mathbb{Z}$ of maximum 0, and that count involves no b . $\square$

Remark 2. The threshold is sharp, and not merely sufficient. Below it the window no longer fits and the identity fails: at $b = 5$, $d = 3$, $n = 3$ the difference $T(6, 3, 3) - T(5, 3, 3)$ is 964,755 while $F(3, 3) = 1,253,329$.

4 Verification

Four checks, each independent of the algebra above.

First, the count $T(4, 1, n)$ was computed by an independent row-pair dynamic program and reproduces the 6 published terms of A125531 that the computation reaches, exactly, in integer arithmetic.

Second, the identity was evaluated directly at the entry’s own alphabet size for every n the threshold covers, and at one n beyond it, where it fails as the remark says.

Third, $F(n, d)$ as computed here agrees with the entry’s own reference sequences. The entry writes them as “A...(*diff* + 1)”, and the index is positional — the (*diff* + 1)st term as listed. That reading is forced: the sequences do not share an offset, and only the positional reading makes $n = 2$ and $n = 3$ agree simultaneously.

Fourth, the following table was produced by the same program and can be checked by hand at small n :

n	$T(4+1, 1, n)$	$T(4, 1, n)$	$F(n, 1)$
1	5	4	1
2	73	54	19
3	5347	3682	1665

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A125531: <https://oeis.org/A125531>.